

IT Project Overview

Crossing the Digital Divide: Network Infrastructure for New Jane Furse Memorial Hospital

Sekhukhune District Municipality, Limpopo | Project Overview Document

1. Scenario Description

The Sekhukhune District Municipality in Limpopo comprises four local municipalities: Elias Motsoaledi, Ephraim Mogale, Fetakgomo Tubatse, and Makhuduthamaga. The district is remote and has historically lagged behind urban centres in network coverage. New Jane Furse Memorial Hospital, the largest public hospital in the district, serves roughly 100 patients daily across 20 wards and 5 administrative offices, supported by 600 staff and only 3 ICT support personnel. The hospital's 15 ageing computers, 3 printers, and largely absent network security controls sit alongside serious concerns about data storage and missing patient files. The hospital is one kilometre from the district municipal building, and Openserve has been identified as the sole carrier able to extend connectivity into the area. The district has commissioned a one-year project to design and roll out network infrastructure that splices from the municipality's existing network to deliver a dedicated, reliable connection to the hospital.

2. Purpose of the Scenario

The purpose of this scenario is to design a network extension solution that closes the digital divide between the Sekhukhune District Municipality and its primary public hospital. Beyond simply provisioning bandwidth, the project must produce an infrastructure that is secure, supportable by a very small local ICT team, and financially sustainable once the implementation partner has left site. The scenario also foregrounds the social and organizational dimensions of network deployment: because a network connects people as much as devices, the solution must be matched to the hospital's actual staffing, skills, and operational capacity, rather than an idealized "best case" deployment that cannot be maintained locally.

3. Preliminary Overview of the IT Solution

The recommended approach is a fibre-optic splice from the Sekhukhune District Municipality's existing network, extended approximately 1 km to the hospital via Openserve as the last-mile carrier. A dedicated fibre link (rather than a shared or best-effort connection) gives the hospital predictable bandwidth for clinical, administrative, and supply-chain systems. On the hospital side, the link terminates at a hardened core switch and firewall, from which a structured cabling and Wi-Fi layer distributes connectivity to the 20 wards and 5 offices. A local server with backup storage addresses the immediate concern over missing patient files, while role-based access control and a basic security stack (firewall, endpoint protection, monitoring) protect the network from external and internal threats. The design deliberately favours proven, low-maintenance equipment over cutting-edge but complex technology, since the hospital will be supported by only three ICT staff after handover.

4. Initial IT Description

Physically, the solution consists of a fibre run from the municipal point of presence to a new server/comms room at the hospital, terminating in a redundant-capable core switch, a next-generation firewall, and a UPS-backed power supply. From there, structured Cat6 cabling and a set of wireless access points cover the wards and offices, with segmented VLANs separating clinical systems, administrative/finance systems, guest and supplier access, and management traffic. The 15 existing computers will be assessed and, where beyond economical repair, replaced; two additional printers will be added to relieve pressure on the current three. A local file server with scheduled off-site/cloud backup is introduced specifically to resolve the data-loss and missing-file concerns raised in the scenario. Remote monitoring tools will alert the 3-person ICT team to outages or performance issues before they affect patient care.

5. Current Solution and Business Process Alignment

Today, the hospital's operations, finance, and supply-chain functions run on disconnected or poorly connected systems, with suppliers accessing Intralink services to coordinate deliveries and stock. Without reliable connectivity, staff likely rely on manual workarounds, paper records, or intermittent access, which directly explains the reported problem of missing patient files and slows supply-chain coordination with external vendors. The new dedicated link directly supports these existing business processes: finance and supply-chain staff gain consistent access to Intralink and related systems, clinical and administrative staff gain a stable platform for electronic patient records, and the ICT team gains the visibility needed to keep these services running. The solution is deliberately built around the hospital's current software and workflows rather than requiring a parallel system change, minimizing disruption during the one-year rollout.

6. Potential Risks

Sustainability risk: a small 3-person ICT team may be unable to maintain a complex network without ongoing vendor or municipal support after handover.

Single-carrier dependency: Openserve is the only available carrier, creating a single point of failure for the last-mile connection.

Security risk: the near-total absence of existing security controls leaves the network vulnerable during the transition period.

Data risk: without a properly implemented backup and access-control regime, the current problem of missing patient files could persist or worsen.

Social/political risk: as with any infrastructure imposed on an existing human network, insufficient buy-in from hospital staff and municipal stakeholders could undermine long-term adoption and maintenance.

Timeline risk: a one-year deadline is tight for permitting, trenching, procurement, installation, and staff training in a remote district.

Aging hardware: 15 old computers may not support modern security or clinical software, limiting the benefit of improved connectivity.

7. Cost Estimations

The table below provides an indicative, high-level cost estimate for the one-year rollout. Figures are preliminary and would be refined following a formal site survey and vendor quotation from Openserve and hardware suppliers.

Cost Item	Description	Estimated Cost (ZAR)
Fibre splice & trenching (1 km, Openserve)	Duct, fibre cable, trenching/road-crossing permits between municipal node and hospital	R 250,000
Network hardware	Core switch, router, media converters, rack, UPS	R 180,000
Wi-Fi & LAN cabling (wards & offices)	Structured cabling, 25 access points across wards/offices	R 220,000
Security infrastructure	Firewall, VPN, endpoint protection, CCTV for server room	R 150,000
Server & storage	Local server, backup NAS, UPS for server room	R 200,000
End-user devices	20 replacement desktops/thin clients, 2 printers	R 160,000
ICT staff training	Installation, troubleshooting and maintenance training for 3 staff	R 60,000
Project management & contingency (10%)	Design, testing, commissioning, 10% contingency	R 122,000
Estimated Total		R 1,342,000